

# Comment on “Potentially singular behavior of the 3D Navier-Stokes equations” (Found. Comput. Math. 23 (2023) 2251-2299)

[Author]  
[Affiliation]

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## Summary

Hou [H23b] exhibits a family of axisymmetric Navier-Stokes solutions with swirl whose amplitude grows like  $(T - t)^{-1}$  at a fitted singular time  $T$ , and in [H24, Sec. 1.5] identifies a logarithmic growth of  $|ru^r|$  as the mechanism by which the scenario would escape the exclusion theorems for axisymmetric Type I singularities with swirl. Those theorems (Koch, Nadirashvili, Seregin and Sverak [KNSS09, Thms. 5.3 and 6.1]; Seregin and Sverak [SS09, Thm. 1.1]) do not involve  $|ru^r|$ ; their hypothesis is a bound on  $|u||x'|$ , which is scale-invariant and does not require the singular time. We report that on a resolved continuation of the first-stage ( $\nu = 5 \times 10^{-4}$ ) solution of [H23b] from 144 to 331 times the initial amplitude,  $\sup_x |u(x, t)||x'|$  is constant to 0.3 percent (72.2), so that the computed scenario satisfies the hypothesis under which the theorems exclude a singularity. A proposition (Section 3) records the implication in the cylinder setting of [H23b]. The accompanying paper [our preprint] contains the computation, its certificates, and the first non-axisymmetric stability analysis of the same solution.

## 1 Setting

Let  $u$  be a smooth solution of the incompressible Navier-Stokes equations with viscosity  $\nu > 0$  on the periodic cylinder  $D = \{(x', x_3) : |x'| < 1, x_3 \in \mathbb{R}/\mathbb{Z}\}$ , axisymmetric with swirl, with the no-slip (or the Hou-Luo) condition at the wall  $|x'| = 1$ , on a time interval  $(0, T)$ , and suppose  $T$  is a first singular time. Write  $r = |x'|$ ,  $u = u_r e_r + u_\theta e_\theta + u_z e_z$ ,  $\Gamma = ru_\theta$ . This is the setting of Hou’s Navier-Stokes scenario [H23b] (with the swirl profile  $12000(1 - r^2)^{18} \sin(2\pi z)/(1 + 12.5 \sin^2(\pi z))$  as initial data and the two-stage viscosity  $5 \times 10^{-4}$  then  $5 \times 10^{-3}$ ).

## 2 The theorems used

**Theorem A (Koch, Nadirashvili, Seregin, Sverak [KNSS09], Theorem 5.3).** A bounded weak axisymmetric solution  $v$  on  $\mathbb{R}^3 \times (-\infty, 0)$  satisfying  $|v(y, s)| \leq C/|y'|$  for all  $(y, s)$  is identically zero.

**Theorem B** ([KNSS09], Theorems 6.1-6.2). An axisymmetric weak solution on  $\mathbb{R}^3 \times (0, T)$  satisfying  $|u(x, t)| \leq C/|x'|$  is bounded (and mild); a Type I solution,  $|u| \leq C/\sqrt{T-t}$ , satisfying  $|u| \leq C/|x'|$  for  $|x'| \geq R_0$  is bounded.

**Theorem C** (Seregin, Sverak [SS09], Theorem 1.1 and Lemma 3.3). For an axisymmetric weak solution in  $L_3$  with pressure in  $L_{3/2}$  on a parabolic cylinder about an axis point, a Type I bound on the meridional velocity alone,  $\sup \sqrt{t_0 - t} |(u_r, u_z)| < \infty$  on the cylinder, gives regularity at that point; singular points of an axisymmetric suitable weak solution lie on the axis.

None of these hypotheses involves  $ru^r$ ,  $ru^z$ , or a log-weighted circulation.

### 3 Proposition (proof by citation)

**Proposition.** In the setting of Section 1, suppose that

$$K := \sup_{0 < t < T} \sup_{x \in D} |u(x, t)| |x'| < \infty. \quad (\text{H})$$

Then  $T$  is not a singular time.

*Proof.* (i) By Theorem C (Lemma 3.3) and the smoothness of  $u$  away from the axis for  $t < T$ , a singular point at time  $T$  lies on the axis; the wall  $|x'| = 1$  is at distance one from the axis and the no-slip condition makes  $u$  smooth up to the wall for  $t < T$ , so the argument is local and the cylinder enters only through the existence of the solution on it. (ii) Let  $x_0$  be an axis point and suppose it is singular at  $T$ . Under (H) the Leray rescaling  $u^{(\lambda)}(y, s) = \lambda u(x_0 + \lambda y, T + \lambda^2 s)$  satisfies  $|u^{(\lambda)}(y, s)| \leq K/|y'|$  for all  $\lambda$  and all  $s < 0$ , uniformly. If the solution is Type I at  $x_0$  ( $|u| \leq M/\sqrt{T-t}$  on a parabolic cylinder about  $x_0$ ), the family is bounded and, by the standard compactness of Type I rescalings (the mild-solution local theory as in [SS09] and [AB19]), a subsequence converges locally uniformly to a bounded ancient axisymmetric solution  $v$  with  $|v(y, s)| \leq K/|y'|$ , normalized so that  $|v|$  is of order one somewhere; Theorem A gives  $v = 0$ , a contradiction. If the solution is not Type I at  $x_0$ , Theorem B (Theorem 6.1, applied on a parabolic cylinder about  $x_0$  where (H) holds; its proof is local) gives that  $u$  is bounded near  $x_0$  up to  $T$ , hence regular there. Either way  $x_0$  is regular. Since every candidate singular point is on the axis and is regular,  $T$  is not a singular time. QED

*Remarks.* (a) The proposition contains no new mathematics; its content is that (H), and not any bound on  $ru^r$ , is what the theorems ask of Hou's scenario. Hou's stated escape [H24, Sec. 1.5], a logarithmic growth of  $|ru^r|$ , is not a hypothesis of Theorems A-C; Barker [B25] makes the same observation. (b) (H) is scale-invariant and does not involve the singular time  $T$ , so it can be measured on a computation without a fitted  $T$ .

### 4 The measurement

On the continuation of Hou's first-stage viscous solution ( $\nu = 5 \times 10^{-4}$ ) from 144 to 331 times the initial swirl amplitude, over which  $\log(1/(T-t))$  rises from 10.6 to 13.9 with  $T$  fitted by the log-derivative method ( $0.0022865 \pm 1 \times 10^{-7}$ ), the quantity  $\sup_x |u(x, t)| |x'|$  reads

$$72.3, 72.4, 72.2, 72.3, 72.3, 72.3, 72.2, 72.2, 72.2, 72.2, 72.1$$

on the twelve resolved snapshots (amplifications 144 to 331), constant to 0.3 percent. The supremum is attained in the outer flow, not in the collapsing core; its value at the swirl maximum itself is 21.1, 22.1, 21.8 at amplifications 144, 217, 268 (20.5 at 331, where the map limits it), flat within its

scatter; the circulation maximum is 64.9 to 64.7 (its maximum principle); the Type I constant  $\sup |u| (|x - x_0| + \sqrt{T - t})$  is 75.4 to 74.6; the meridional Type I constant  $\sup |(u_r, u_z)|\sqrt{T - t}$  is 6.0 to 7.7. A logarithmic growth of  $|u|r$  over this range would read about thirty percent. The front of the collapse is resolved to ten cells on a front-following map and is the Rott-Lundgren strained layer under the measured collapse strain to a five percent profile residual; the amplitude exponent is  $1.03 \pm 0.03$ .

**Conclusion.** As computed, the scenario satisfies (H) with  $K = 72$  through the whole resolved range, with no trend, the bound carried by the outer flow and the core value flat. The only way for it to be singular is for  $K$  to become infinite as  $t \rightarrow T$ ; a core quantity growing logarithmically from 21 would reach the outer value only after a factor of about three, so the data exclude such growth over the resolved range at the core (a 30 percent rise would have shown) but cannot exclude it in the limit, and on the core-restricted  $|ru^r|$  that Hou names the data are inconclusive (non-monotone between 15 and 22 over the range: a 30 percent fall to 72 times the initial amplitude, a 45 percent rise to 266 times, and a 13 percent disagreement between maps at the end; and across the late-stage transition at about 400 times the initial amplitude, where the one-scale collapse ends and a second structure nearer the symmetry plane takes over, it rises from 16.5 to 20.7 in a third of a log-unit, a jump between regimes rather than a trend). A finite computation cannot establish (H) up to  $T$ ; it can show the absence of the departure the scenario needs over the range computed, which it does. Whether a logarithmic growth of  $K$  could be excluded by theorem is open: the KNSS argument uses (H) uniformly over rescalings and fails structurally under any logarithmic loss, and the sharpest growth-side results (Pan [P16]; Seregin [S21]) reach only a double logarithm to a small power. That question, which reduces to the dependence of the axis Holder exponent in the critical-drift oscillation decay on the drift constant, is the one analytical question this scenario poses.

## References

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